

Roll No.

TMA-101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Nov., 2022

ENGINEERING MATHEMATICS—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Find rank of the matrix :

(CO1)

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

OR

(b) Determine whether the following set of vectors are linearly dependent

or independent : $\{(0, 1, 1), (1, 0, 3), (1, 1, 4)\}$.

(CO1)

P. T. O.

2. (a) For what value of λ and μ , the following system of equations (CO2)

$$x + y + z = 6;$$

$$x + 2y + 3z = 10;$$

$$x + 2y + \lambda z = \mu$$

will have :

(i) Unique solution;

(ii) Infinite solutions;

(iii) No solution.

OR

- (b) Diagonalize the matrix : (CO2)

$$\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

3. (a) Find Eigen values and Eigen vectors of the matrix : (CO1)

$$\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$

OR

- (b) If $y = \sin(m \sin^{-1} x)$, prove that : (CO3)

$$(1 - x^2)y_{n+2} - (2n + 1)x.y_{n+1} + (m^2 - n^2)y_n = 0.$$

(3)

4. (a) If $u = \log(x^3 + y^3 + z^3) - 3xyz$, then show that : (CO3)

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = -\frac{9}{(x + y + z)^2}.$$

OR

- (b) If $u = \log_e \left(\frac{x^3 + y^3}{x + y} \right)$, show that : (CO3)

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2.$$

5. (a) Expand $e^x \cos y$ near the point $(0, 0)$ by Taylor's theorem upto third degree term. (CO3)

OR

- (b) Expand $x^2y + 3y - 2$ in powers of $x - 1$ and $y + 2$ using Taylor's theorem. (CO3)

Roll No.

EEEC-101**B. TECH. (BIOTECHNOLOGY) (FIRST SEMESTER)****MID SEMESTER EXAMINATION, Nov., 2022****BASIC ELECTRICAL AND ELECTRONICS ENGINEERING****Time : 1½ Hours****Maximum Marks : 50**

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the types of semiconductors. (CO1)

OR

- (b) Explain with the help of energy band diagram for metals, insulators and semi-conductors. (CO1)

2. (a) State the Norton's theorem. Find out current in 5Ω resistor with the help of Norton's Theorem in the circuit shown in Fig. 1. (CO1 & CO2)

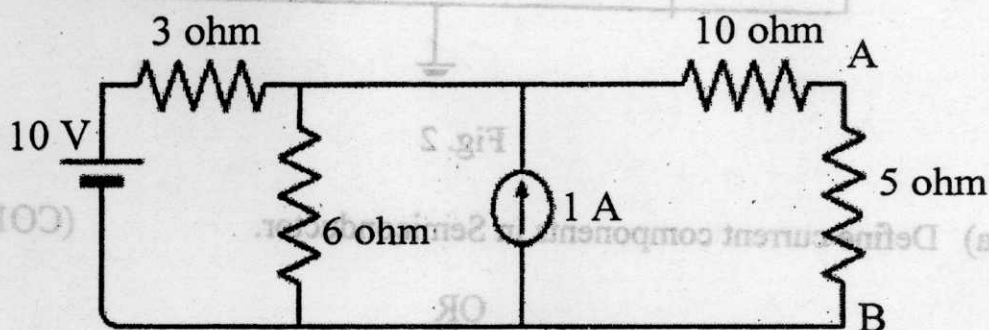


Fig. 1

P. T. O.

OR

- (b) Define Mass Action Law. Derive an expression for charge densities of semiconductor materials. (CO1 & CO2)

3. (a) What are star and delta configurations ? Derive the expressions for star to delta and delta to star transformations in a two terminal dc circuit.

(CO1)

OR

- (b) Find out the voltages at nodes 1 and 2 using Nodal analysis as shown in Fig. 2. (CO1)

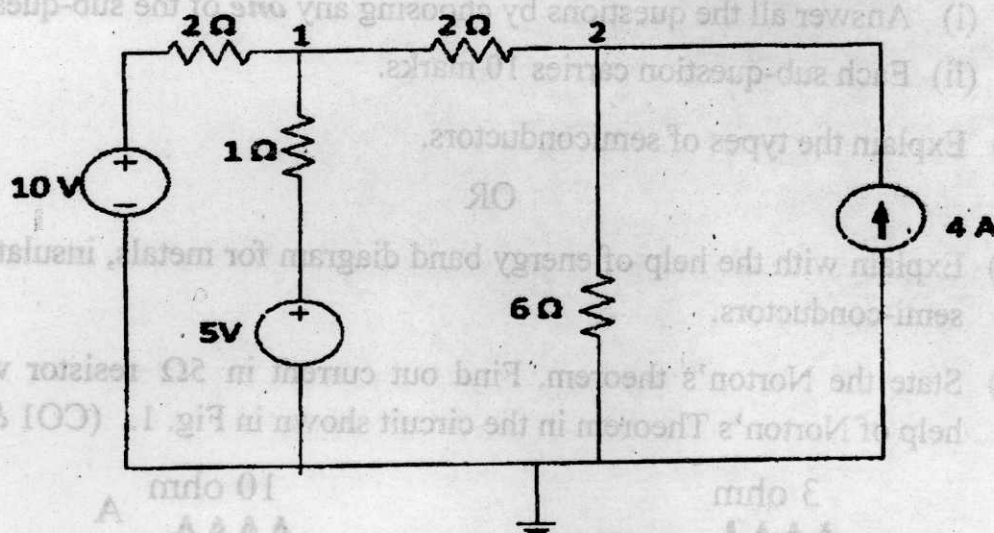


Fig. 2

4. (a) Define current components in Semiconductor. (CO1 & CO2)

OR

- (b) Differentiate between avalanche and Zener breakdown. (CO1 & CO2)

(3)

5. (a) Write down the statement of maximum power transfer theorem ? Derive the condition for maximum power transfer from source to load in DC Circuit. Also derive the expression of maximum efficiency in maximum power transfer condition in dc circuit. (CO1)

OR

- (b) Define the following terms along with suitable examples : (CO1)
- (i) Unilateral and Bilateral Elements
 - (ii) Active and Passive Elements
 - (iii) Linear and Non-linear elements
 - (iv) KCL and KVL

TEE-101

Roll No.

B. TECH. (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Nov., 2022
BASIC ELECTRICAL ENGINEERING
Time : 1½ Hours
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define the following AC terms :

(CO1 & CO2)

- (i) RMS value
- (ii) Peak value
- (iii) Form factor
- (iv) Peak factor
- (v) Instantaneous value

OR

(b) Find out the voltages at nodes 1 and 2 using Nodal analysis as shown in Fig. 1 :

(CO1 & CO2)

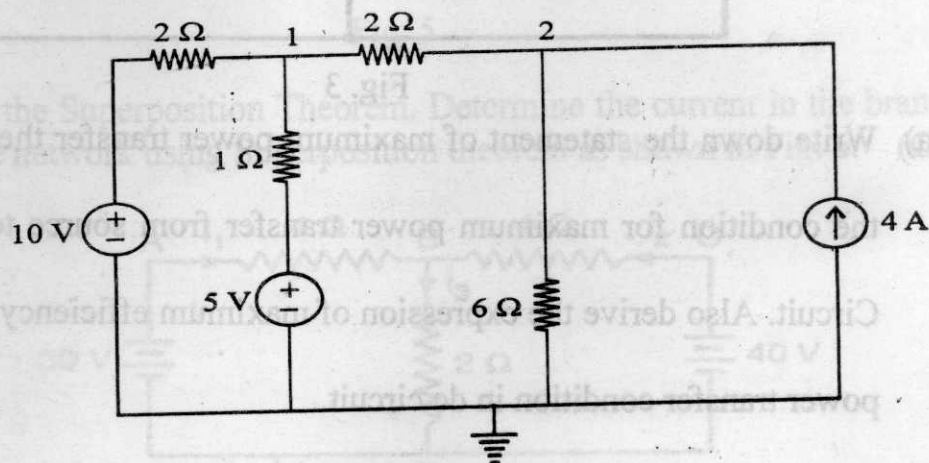


Fig. 1

P. T. O.

2. (a) State the Norton's theorem. Use source transformation technique to find the load current 'I' in the circuit shown in Fig. 2 : (CO1)

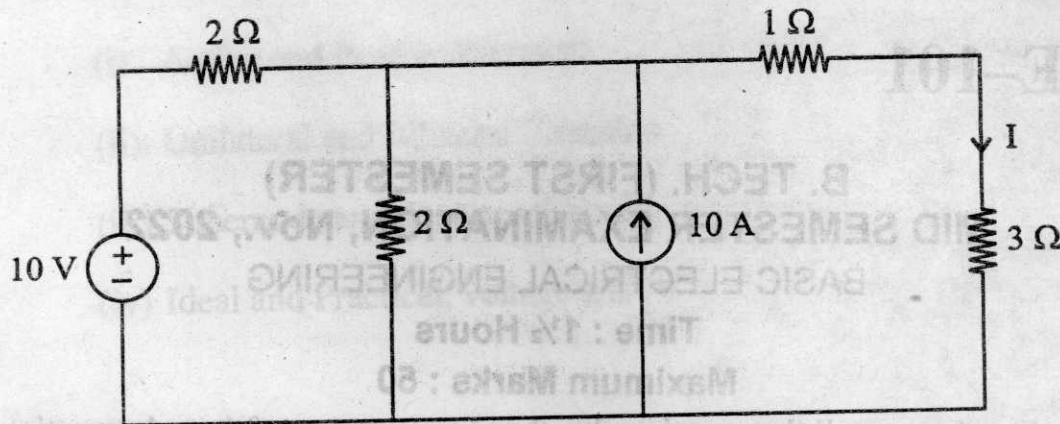


Fig. 2

OR

- (b) Write down all the required steps for Mesh analysis. Find the current across 2Ω resistor as shown in Fig. 3 by using mesh analysis. (CO1)

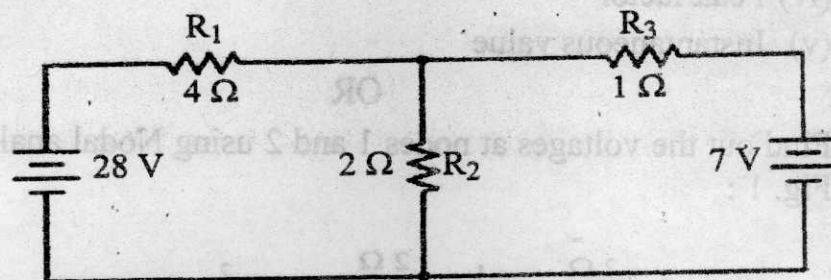


Fig. 3

3. (a) Write down the statement of maximum power transfer theorem. Derive the condition for maximum power transfer from source to load in DC Circuit. Also derive the expression of maximum efficiency in maximum power transfer condition in dc circuit. (CO1)

OR

- (b) Determine the current through 15Ω resistor using Thevenin's theorem as shown in Fig. 4 : (CO1)

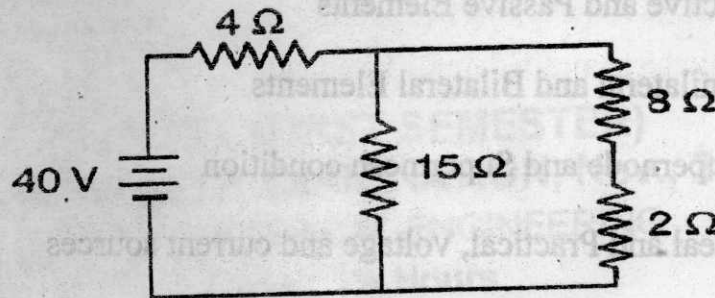


Fig. 4

4. (a) What are star and delta configurations ? Derive the expressions for star to delta and delta to star transformations in a two terminal dc circuit. (CO1)

OR

- (b) Calculate current through 20Ω resistor using Norton's theorem in the circuit shown in Fig. 5 : (CO1)

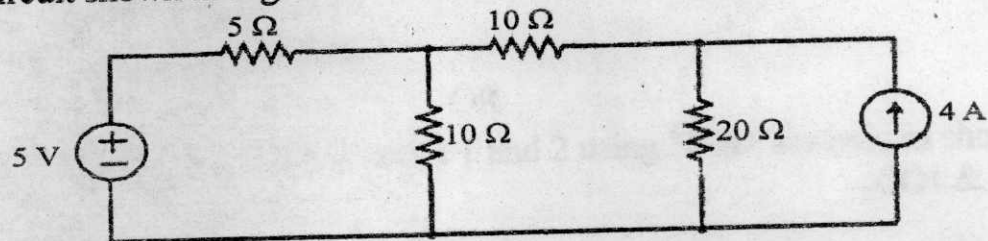


Fig. 5

5. (a) State the Superposition Theorem. Determine the current in the branches of the network using Superposition theorem as shown in Fig. 6. (CO1)

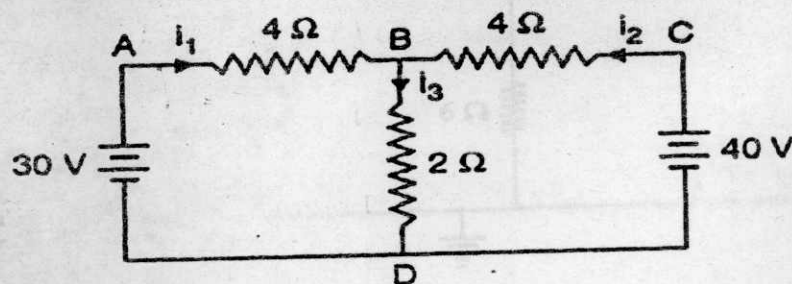


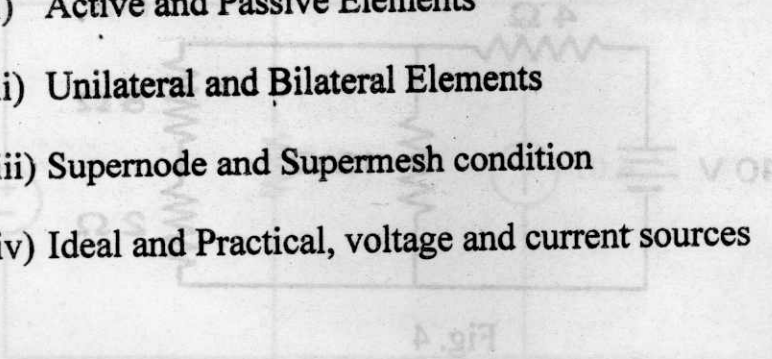
Fig. 6

P. T. O.

OR

(b) Define the following terms along with suitable examples : (CO1)

- (i) Active and Passive Elements
- (ii) Unilateral and Bilateral Elements
- (iii) Supernode and Supermesh condition
- (iv) Ideal and Practical, voltage and current sources

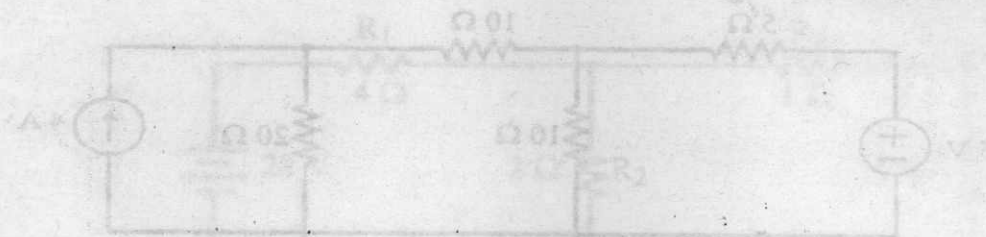


4. (a) What are star and delta configurations? Derive the expressions for star to delta and delta to star transformations in a two terminal dc circuit.

(CO1)

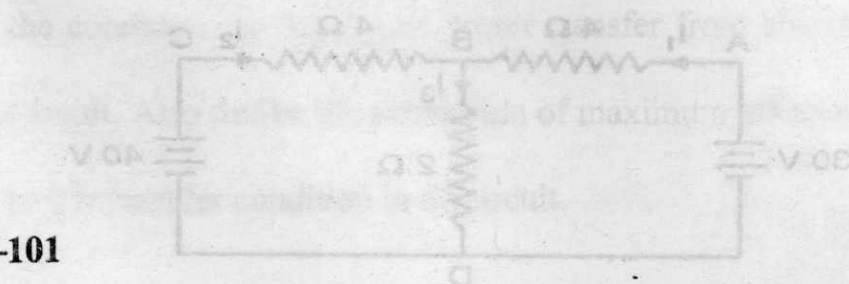
(b) Calculate current through 20Ω resistor using Norton's theorem in the circuit shown in Fig. 5 :

(CO1)



5. (a) State the Superposition Theorem. Determine the current in the branches of the network using superposition theorem as shown in Fig. 6. (CO1)

(CO1)



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TEC-101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Nov., 2022

BASIC ELECTRONICS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Convert the number : (CO1)

(i) $\sqrt{(41)_x} = (5)_x$, find the base "x".

(ii) $(3213)_4 = (?)_{10}$

(iii) $(613263)_8 = (?)_{16}$

Simplify the following expressions with the help of Boolean algebra :

(iv) $F = A [B + C (AB + AC)]$

(v) $F = A + AB + AB'C$

OR

(b) Simplify following expression with the help of Boolean algebra : (CO1)

(i) $F = A'BC + AB'C' + A'B'C' + AB'C + ABC$

(ii) $F = AB + (AC)' + AB'C (AB + C)$

(iii) $F = A'B'C + (A + B + C)' + A'B'C'D$

P. T. O.

Convert the following numbers :

(iv) $(823.45)_{10} = (?)_7$

(v) $(A8C2.0A)_{16} = (?)_{10}$

2. (a) Simplify the following function with the help of K Map : (CO1)

$$F(A, B, C, D) = A + BC' + ABD' + ABCD$$

OR

- (b) Minimize the following by K map and realize the minimized function by : (CO1)

(i) NAND gate only

(ii) NOR gate only

$$F(A, B, C, D) = \pi M(0, 1, 3, 5, 7, 8, 9, 11, 13, 14, 15)$$

3. (a) If N_D and N_A are donor and acceptor impurities and n_i is the intrinsic concentration, establish the relation for minority and majority charge densities in extrinsic semiconductors. (CO2)

OR

- (b) Define mobility and drift velocity. Prove that conductivity of a conductor is given $\sigma' = \mu ne$. What would be the conductivity of intrinsic and extrinsic (p-type and n-type) semiconductor ? (CO2)

4. (a) A bar of pure silicon has cross sectional area of 2 mm^2 and intrinsic concentration of silicon is $2.5 \times 10^{16} \text{ m}^{-3}$. The free electron and hole motilities are $0.13 \text{ m}^2/\text{V-sec}$ and $0.05 \text{ m}^2/\text{V-sec}$ respectively. Find the conductivity and length of bar whose resistance is $50 \text{ k}\Omega$. (CO2)

(3)

OR

- (b) In a doped semiconductor, there are 4.52×10^{24} holes/m³ and 1.25×10^{14} electrons/m³. What will be the carrier density of undoped specimen ?

If electron and hole motilities are $0.38 \text{ m}^2\text{V}^{-1}\text{sec}^{-1}$ and $0.18 \text{ m}^2\text{V}^{-1}\text{sec}^{-1}$ respectively, determine conductivity of intrinsic and doped semiconductor. (CO2)

5. (a) A small concentration of minority carries is injected into a homogeneous semiconductor crystal at one point. An electric field of 10 V/cm is applied across the crystal and this moves the minority carries a distance of 1 cm in $20 \mu \text{ sec}$. Determine mobility (in $\text{cm}^2/\text{V-sec}$).

(CO1/CO2)

OR

- (b) Simplify the following expression and realize the minimized function by NOR gate only : (CO1/CO2)

$$F(A, B, C) = A'B'C + A'BC' + A'B + B'C'$$

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TPH-101

**B. TECH. (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Nov., 2022**

ENGINEERING PHYSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-questions carries 10 marks.

1. (a) In an Young's double slit experiment, derive the expression for the fringe width of the interference pattern. (CO1)

OR

- (b) In a typical biprism experiment $b/a = 20$, and for sodium light ($\lambda = 5893 \text{ \AA}$) one obtains fringe width of 0.1 cm; here b is the distance between the biprism and the screen, a is the distance between virtual source to the biprism. Assuming $\mu = 1.5$, calculate the prism angle α .

2. (a) What are the Newton's rings ? Describe the conditions of formation of the Newton's rings. (CO1)

OR

- (b) In a Newton's rings set up, diameter of 18th dark ring is found to be 9.5 mm. The space between spherical surface and the flat slab is then filled with water ($\mu = 1.33$). Calculate the diameter of the 14th dark ring in new setup.

P. T. O.

(2)

3. (a) Derive an expression for the intensity distribution due to Fraunhofer diffraction at a single slit and discuss the condition for principal maxima and minima. (CO1)

OR

- (b) In a diffraction grating the sodium doublet ($5890 \text{ \AA}$, $5896 \text{ \AA}$) is viewed in fourth order at 30° to the normal and is just resolved. Determine the grating element and the total width of the rulings.
4. (a) Discuss about the resolving power of the N-slit diffraction grating. (CO1)

OR

- (b) Write down the expression for intensity in N-slit Fraunhofer diffraction and hence obtain conditions for principal maxima and minima.
5. (a) Discuss the types of fiber optics. What is Numerical Aperture (NA); also discuss its significance. (CO2)

OR

- (b) An optical fiber has a numerical aperture of 0.25 in air and cladding refractive index of 1.5. Find the numerical aperture of the fiber in a liquid of refractive index 1.3.

Roll No.

TCS-101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Nov., 2022

**FUNDAMENTAL OF COMPUTER AND INTRODUCTION TO
C PROGRAMMING**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain Von Neumann architecture of computer with a neat diagram.

(CO1)

OR

(b) Define the following :

(CO1)

(i) Utility software

(ii) Cache memory

(iii) Tracks and sectors

(iv) Mesh and ring topology

P. T. O.

2. (a) Find the following output with explanation : (CO1, CO2, CO3, CO4)

(i) #include<stdio.h>

```
int main( )
```

```
{
```

```
    int k = 0;
```

```
    for(k)
```

```
        printf("Hello");
```

```
    return 0;
```

```
}
```

(ii) #include<stdio.h>

```
int main( )
```

```
{
```

```
    int a = -5;
```

```
    if(!(a*-1))
```

```
        printf("success");
```

```
    else
```

```
        printf("Failure");
```

```
    return 0;
```

```
}
```

(iii) #include<stdio.h>

```
int main( )
```

```
{
```

```
    int i=5; j=3, k=-2, z;
```

```
    z= !(i&&j || k);
```

```
    printf("%d",z);
```

```
    return 0;
```

```
}
```

(iv) #include <stdio.h>

int main()

{

int z;

z = 4*6/3-12*5.2%4-8;

printf("%d", z);

return 0;

}

(v) 6 GB = bits.

OR

(b) Find the following output with explanation : (CO1, CO2, CO3, CO4)

(i) #include <stdio.h>

int main()

{

int a=0;

if(a=1)

printf("You won");

else

print("You loose");

return 0;

}

P. T. O.

(ii) #include <stdio.h>

int main()

{

int a=5;

switch(a)

{

case 0 :

printf("0");

case 3 :

printf("3");

case 5 :

printf("5");

default:

printf("RABBIT");

}

(iii) #include <stdio.h>

int main()

{

int i=5;

while(i>0)

printf("%d",i);

i--;

return 0;

}

(iv) #include <stdio.h>

```
int main( )
```

```
{
```

```
int i=5, j=3, k=-2, l=5, z;
```

```
z = --i * --j * --l * ++k;
```

```
printf("%d", z);
```

```
return 0;
```

```
}
```

(v) 32 MB = nibble.

3. (a) Explain different phases involved in compilation of a C program. (CO2)

OR

(b) What do you mean by unary, binary and ternary operators ? Explain operator precedence and associativity. Apply it in the following expression and calculate : (CO2)

(i) $8 + 3 - 6 * 7 / 4 > 7 / 3 * 5$

(ii) $5 * 7 + 3 \& \& 9 < 10 - 8 * 4$

4. (a) Draw a flowchart to input two positive integer numbers, calculate the difference of their squares and square of their difference. Print the largest of two results obtained. (CO3)

OR

(b) Draw a flowchart to input a positive integer number then print the sum of cubes of the factors of that number. For example : (CO3)

$N = 12$

$\text{Sum} = 1^3 + 2^3 + 3^3 + 4^3 + 6^3 = 316$

5. (a) Write a C program to calculate final train fare after applying the following conditions : (CO4)

- (i) If passenger is male/female and age is ≥ 6 , then discount = 50%.
- (ii) If passenger is female and age between 12 to 65, then discount = 20%.
- (iii) If passenger age is between 5 to 12, then discount = 75%.
- (iv) If passenger age is less than 5, then discount = 100%.

OR

(b) Write a C program to calculate the sum of the following series : (CO4)

$$10 + x + x^2 + x^3 + x^4 + \dots + x^n$$

Roll No.

TCH-101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Nov., 2022

ENGINEERING CHEMISTRY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-questions carries 10 marks.

1. (a) On the basis of hydrogen bonding, explain the following :

(i) Ice floats on water.

(CO1)

(ii) Orthonitro phenol can be steam distilled while paranitrophenol cannot.

OR

(b) What is meant by bond order ? Calculate the bond order of He_2^+ and O_2 and also explain the magnetic nature.

2. (a) Explain the following terms :

(CO1)

(i) Spectroscopy and its advantages

P. T. O.

- (ii) Electromagnetic spectrum
- (iii) Beer's Lambert law of absorption of light
- (iv) Range of IR Spectroscopy

OR

- (b) Why Infra-red Spectroscopy is known as vibrational spectroscopy ?
Explain different types of vibrations involved in IR Spectroscopy.
3. (a) Describe the principle of Lime Soda Process with the help of chemical reactions and also differentiate between cold lime soda process and hot lime soda process. (CO2)

OR

- (b) Calculate the amount of lime (92% pure) and soda (98% pure) required for treatment of 30,000 litres of water containing : $\text{Mg}(\text{HCO}_3)_2 = 36.5$, $\text{CaCl}_2 = 27.75$, $\text{Ca}(\text{HCO}_3)_2 = 40.5$, $\text{CaSO}_4 = 34.0$, $\text{MgSO}_4 = 30.0$ and $\text{NaCl} = 50$. (CO2)
4. (a) Distinguish between : (CO2)
- (i) Boiler scales and sludges
 - (ii) Temporary hardness and Permanent hardness

OR

- (b) A completely exhausted zeolite softener requires 120 litres of NaCl solution having 100 g/l of NaCl. How many litres of water having hardness 500 ppm can be softened by the zeolite softener.

(3)

5. (a) Describe the construction of Bomb Calorimeter. How is the calorific value determined with the help of Bomb calorimeter ? (CO4)

OR

- (b) When 0.994 g of coal was burnt completely in Bomb calorimeter, the increase in temp. of 2592 grams of water was 2.732°C , if the water equivalent of calorimeter is 3940 g. Calculate GCV and NCV of the fuel. Given : %H in fuel = 8.0, Latent heat of condensation of steam = 587 cal/g.

Roll No.

TMA-101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Nov., 2022

ENGINEERING MATHEMATICS--I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Find rank of the matrix :

(CO1)

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

OR

(b) Determine whether the following set of vectors are linearly dependent

or independent : $\{(0, 1, 1), (1, 0, 3), (1, 1, 4)\}$.

(CO1)

P. T. O.

2. (a) For what value of λ and μ , the following system of equations (CO2)

$$x + y + z = 6;$$

$$x + 2y + 3z = 10;$$

$$x + 2y + \lambda z = \mu$$

will have :

- (i) Unique solution;
- (ii) Infinite solutions;
- (iii) No solution.

OR

- (b) Diagonalize the matrix : (CO2)

$$\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

3. (a) Find Eigen values and Eigen vectors of the matrix : (CO1)

$$\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$

OR

- (b) If $y = \sin(m \sin^{-1} x)$, prove that : (CO3)

$$(1 - x^2)y_{n+2} - (2n + 1)x.y_{n+1} + (m^2 - n^2)y_n = 0.$$

(3)

4. (a) If $u = \log(x^3 + y^3 + z^3) - 3xyz$, then show that : (CO3)

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = -\frac{9}{(x+y+z)^2}.$$

OR

- (b) If $u = \log_e \left(\frac{x^3 + y^3}{x+y} \right)$, show that : (CO3)

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2.$$

5. (a) Expand $e^x \cos y$ near the point $(0, 0)$ by Taylor's theorem upto third degree term. (CO3)

OR

- (b) Expand $x^2y + 3y - 2$ in powers of $x-1$ and $y+2$ using Taylor's theorem. (CO3)

Roll No.

EEEC-101**B. TECH. (BIOTECHNOLOGY) (FIRST SEMESTER)****MID SEMESTER EXAMINATION, Nov., 2022****BASIC ELECTRICAL AND ELECTRONICS ENGINEERING****Time : 1½ Hours****Maximum Marks : 50**

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
 (ii) Each sub-question carries 10 marks.

1. (a) Explain the types of semiconductors. (CO1)

OR

(b) Explain with the help of energy band diagram for metals, insulators and semi-conductors. (CO1)

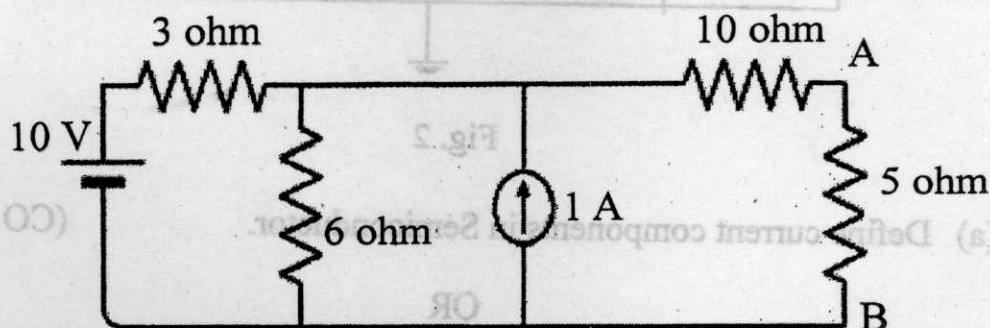
2. (a) State the Norton's theorem. Find out current in 5Ω resistor with the help of Norton's Theorem in the circuit shown in Fig. 1. (CO1 & CO2)

Fig. 1

P. T. O.

OR

(b) Define Mass Action Law. Derive an expression for charge densities of semiconductor materials. (CO1 & CO2)

3. (a) What are star and delta configurations ? Derive the expressions for star to delta and delta to star transformations in a two terminal dc circuit.

(CO1)

OR

(b) Find out the voltages at nodes 1 and 2 using Nodal analysis as shown in Fig. 2. (CO1)

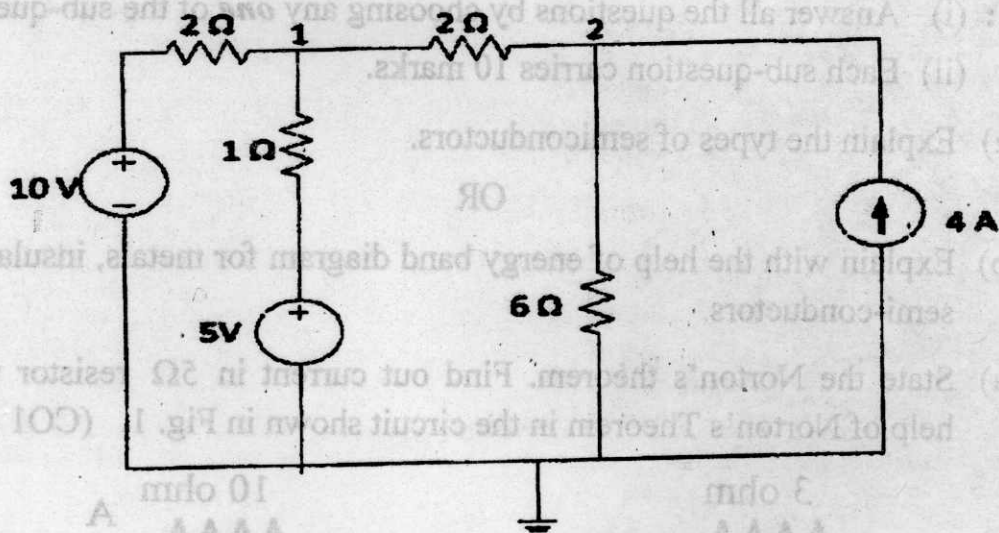


Fig. 2

4. (a) Define current components in Semiconductor.

(CO1 & CO2)

OR

(b) Differentiate between avalanche and Zener breakdown. (CO1 & CO2)

(3)

5. (a) Write down the statement of maximum power transfer theorem ? Derive the condition for maximum power transfer from source to load in DC Circuit. Also derive the expression of maximum efficiency in maximum power transfer condition in dc circuit. (CO1)

OR

- (b) Define the following terms along with suitable examples : (CO1)

- (i) Unilateral and Bilateral Elements
- (ii) Active and Passive Elements
- (iii) Linear and Non-linear elements
- (iv) KCL and KVL

Roll No.

TCE-101

**B. TECH. (CE) (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Nov., 2022**

BASICS OF CIVIL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain briefly the history of civil engineering in India. (CO1)

OR

(b) Discuss about the scopes of civil engineering. (CO1)

2. (a) Write in details about any *two* branches of civil engineering. (CO1)

OR

(b) What are the factors to be considered while selecting a site for construction of residential building ? (CO1)

3. (a) How civil engineers play a major role in infrastructural development of nation ? (CO1, CO2)

OR

(b) Explain the fundamental principles of surveying. (CO1, CO2)

P. T. O.

(2)

4. (a) Discuss the classification of surveying based on instruments used. (CO2)

OR

- (b) Distinguish between plans and maps. Define representative fraction. (CO2)

5. (a) How bearing is used to calculate the angular measurement in surveying? (CO2)

OR

- (b) Explain the process of remote sensing and its uses in surveying. (CO2)